

Siddhartha Devulapalli

AI/ML Engineer

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Skills

Languages: Python, SQL, Java, JavaScript, TypeScript, HTML/CSS

Frameworks: FastAPI, Flask, Django, React, Next.js, Node.js

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, SciPy, Hugging Face (Transformers/Diffusers)

Gen AI & LLMs: RAG, LangChain, LangGraph, LlamaIndex, CrewAI, Prompt Engineering, Vector Databases (Pinecone, ChromaDB, FAISS)

Data Engineering: NumPy, Pandas, Matplotlib, Seaborn, Apache Spark, ETL Pipelines

MLOps & Cloud: Git, Linux, Docker, Kubernetes, MLflow, LangSmith, Arize Phoenix, CI/CD, AWS (SageMaker), GCP (Vertex AI), Azure (AI Studio/Databricks)

Mathematics: Linear Algebra, Calculus, Statistics & Probability, Optimization Techniques

Experience

AI/ML Engineer

Aztra.ai

Hyderabad, Telangana

Aug 2025 – Present

- **Aurora (Demand Forecasting):** Developed a retail demand sensing engine using **Azure Databricks** and **Nixtla**, achieving a **97% Forecast Accuracy (WAPE)** and **90% Forecast Bias**.
- Built **Promo Uplift** and **Causal ML** models (X-learners) to quantify marketing impact, integrating exogenous variables like weather and social pulses for a **24–48 hour feedback loop**.
- Architected a high-level solution using **FastAPI** and **Azure Cache for Redis** to serve store-level granular forecasts for dynamic inventory allocation.
- **Aries (LLM Automated Testing tool):** Engineered a hybrid **GraphRAG** system using **Neo4j** and **FAISS** to automate codebase debugging and test case generation via **OpenAI Embeddings**.
- Orchestrated complex AI workflows using **AWS Step Functions** and **LangGraph**, automating synthetic data creation and incident reporting for **Jira** and **ServiceNow**.
- Implemented **LLM-as-a-Judge** frameworks and prompt guardrails to evaluate API signatures and detect anomalies in server logs with **Claude 4.5 Sonnet**.

Business Analyst

Darwinbox

Hyderabad, Telangana

Jun 2021 – Mar 2023

- Streamlined contract validation processes, saving **\$5K annually**.
- Led quarterly deal audits, securing deals worth **\$50-100K per quarter**.
- Managed AWS credits, saving \$3-5K quarterly and redeeming **\$35K annually**.
- Negotiated vendor contracts, reducing costs by **\$10-15K annually**.
- Contributed to Tableau dashboards for metric reporting presented to co-founders.

Projects

🔗 LLM-based Automation Agent

An automation agent leveraging LLMs to process tasks like log analysis and API integration.

Python, FastAPI, Docker, GPT-4o-Mini

Jan 2025 – Feb 2025

- Built an API with endpoints to execute plain-English tasks and verify outputs against expected results.
- Automated tasks such as formatting files, sorting JSON arrays, and extracting key information using LLM function calling.
- Developed a **Dockerized CI/CD pipeline** for deployment; published Docker images publicly on Docker Hub.

🚗 Car Price Prediction App

An ML-powered app to predict car prices based on input features using regression models.

Python, Streamlit, Scikit-learn

Feb 2024 – Mar 2024

- Built an ML regression model with a **92% R² score** using over **20K training points**.
- Deployed the app online using **GitHub** and **Streamlit Deployment**.

Grocery Store App (Full-Stack)

Python, Flask, Vue.js, SQLite, Bootstrap, YAML

*A full-stack grocery store application with role-based access control for **CRUD** operations.*

July 2023 – Oct 2023

- Developed back-end using **Flask** frameworks and front-end using **Vue.js** for a responsive UI.
- Implemented role-based access control: Admin for managing categories/products and User for purchasing items.
- Documented APIs using **YAML** and deployed the app via GitHub.

Publications

"Evaluating Harmful Instructions Compliance in LLMs: From Circuit Breakers to Multi-Turn Attacks", Siddhartha Devulapalli, Manuscript in Preparation for AI Safety Workshop. 2026

Developed a modern evaluation framework to quantify safety-utility tradeoffs in LLMs (Llama-3.1, Mistral) using StrongREJECT metrics; engineered a pipeline to analyze linear safety degradation in multi-turn attacks and the efficacy of inference-time defenses like SafeDecoding.

Education

IIT Madras

Chennai, Tamil Nadu

B.S in Data Science and Applications, CGPA: 9.8

2022 - 2026

ICFAI Business School

Hyderabad, Telangana

BBA in Business Analytics, CGPA: 10

2018 - 2021

Certifications

Scaler Data Science and Machine Learning, Google Cloud Computing Foundations, Microsoft Azure AI Fundamentals, Develop GenAI Apps with Gemini and Streamlit, Introduction to Data Engineering using Azure, Fundamentals of NLP

Achievements

- **Top 1%** academic performer at IIT Madras; Awarded **Certificate of Academic Distinction**
- Contributed to **LLM Research** for higher education as a Teaching Assistant at IIT Madras.
- Awarded **Gold Medal** and a cash prize of **INR 30K** for academic excellence (1st Rank) at ICFAI Business School.